Fast Deterministic PDF Text Reconstruction

Abstract — We present an offline document extraction pipeline for modern desktop environments. The method combines PDFium native stream parsing with local ONNX-based OCR routing.

1. Introduction

Document conversion in privacy-critical environments requires fully local computation.
Cloud-based extraction introduces telemetry risks and latency penalties.

2. Methodology

Our pipeline classifies pages into born-digital text and raster scans.
Digital pages bypass rasterization completely, achieving sub-millisecond throughput.

3. Experimental Results

Evaluation across diverse benchmarks demonstrates 0.0% CER on born-digital text.
Peak heap memory remains below 10 KiB for native extraction pipelines.

4. Conclusion

Offline document conversion provides superior privacy and predictable resource limits.